

A Jai-inspired systems language in Rust. Incremental salsa front end, lossless CST, typed SSA mid-end, two execution engines held byte-identical by a differential harness.

6/6 gates green 981 tests ADR-0097 latest W5 COMPLETE · 15 sub-waves

| TESTS                  | CORPUS                 | ADRS                    | DIAGNOSTICS            | EDITOR CHECKS              |
|------------------------|------------------------|-------------------------|------------------------|----------------------------|
| 981                    | 193                    | 97                      | 107                    | 166                        |
| workspace, all passing | jr files, both engines | 0001 to 0097, immutable | codes, E0277 next free | Neovim, verified not gated |

The vertical slice, and what it costs to be honest about it

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| <p>EXIT CRITERIA</p> <p><b>DONE</b> hello.jr runs in the VM</p> <p><b>DONE</b> compiles to a native binary</p> <p><b>DONE</b> rustc-grade diagnostics</p> <p><b>DONE</b> formatter round-trips the corpus</p> <p><b>DONE</b> editor integration (Neovim)</p> <p><b>OPEN</b> verified Linux x86-64 CI run</p> <p>The last one needs a push. Configured, never run, so it is a decision rather than a technical gap.</p> | <p>HOW CORRECTNESS IS ESTABLISHED</p> <p><b>Both engines run every corpus program</b> and must agree, on output and on trap wording, from one shared formatter.</p> <p><b>A plan's stated reason is checkable.</b> W4.5 was scheduled after W4 because exhaustiveness "wants comptime type info". Three greps and a two-line program showed it does not: the enum member set is already in the pool during checking. The wave moved forward and the table records the amendment — a wave order resting on a dependency that does not exist is a plan contradicting itself.</p> <p><b>Check the query a claim is about.</b> "The arithmetic around a #run is not folded" was true of the <b>built</b> MIR and false of the <b>optimized</b> one — and the corpus only ever snapshots the built query, so nothing the tests display would have shown it. A claim about optimisation needs a probe of the optimized body.</p> <p><b>A well-typed placeholder is invisible to every gate this project has.</b> <code>t := Point; —</code> a type bound to a local — type-checked cleanly and <b>both engines exited 0</b>, storing an undefined value into a slot of a type with no runtime layout at all. Three of these now, and each was found by asking a question no test asks: what does this lower to? The corpus checks exit codes and snapshots the built MIR; a construct that is legal, silent and unread sits outside all of it.</p> <p><b>A test naming an unimplemented thing has a one-wave shelf life.</b> The refused-body test has now had its construct replaced twice, both times because the wave after it implemented the gap it named. It uses something refused <b>by design</b> now. And one of this wave's own new tests passed vacuously until it was teeth-checked.</p> |
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The language today

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| <p>WORKS, END TO END</p> <p>Full integer tower, bool, string, pointers</p> <p>float32 and float64, plain IEEE-754, no traps</p> <p>struct, nominal, one level</p> <p>union, nominal, untagged — a cross-field read reinterprets</p> <p>variant — a tagged union: a wrong-case read traps, switch destructures</p> <p>enum and enum_flags, namespaced, bare dot-member, switch cases</p> <p>Fixed arrays and views, bounds-checked; a length may name a constant</p> <p>cast and xx from context; operator overloading</p> <p>Trapping arithmetic, wrapping variants, bitwise</p> <p>if, else, while, for, break, continue, defer, using</p> <p>switch with exhaustiveness checking over an enum; else</p> <p>Multiple returns, named args, literal defaults</p> <p>import, foreign, system_library, #scope_module</p> <p>\$T procedures, Box(\$T) structs, and \$N comptime-value parameters —</p> <p>complete, including [N]T sized by a \$N; all instantiated in both engines</p> <p>Compile-time run at file scope or in a body, across files; type_info(), Any, #insert,</p> <p>#code</p> <p>A type as a compile-time value: <code>T :: Point</code> aliases one, usable anywhere <code>Point</code> is</p> <p>A type in a runtime position is refused — it has no representation to store</p> <p>Traps name their source line and the live call chain</p> <p>Bounds checks, strippable by build setting or <code>#no_abc</code></p> <p>context, a hidden parameter passed by pointer</p> <p>Procedures as values: call, pass, return, struct field</p> <p>null, a context-typed pointer literal; malloc and free</p> <p>An allocator in the context, installed and called through</p> <p>Pointer offset <code>p + n</code>, <code>n + p</code>, <code>p - n</code> — element-scaled, unchecked</p> <p>push_context, a block with its own copy of the context</p> <p>talloc / reset_temporary_storage — a per-context bump arena</p> | <p>ABSENT (WAVE)</p> <p>pointer difference <code>p - q</code>, deferred</p> <p>percent on floats, <code>is_nan</code>, math (W7)</p> <p>a recursive variant; eliding the check in an arm</p> <p>an explicit backing type</p> <p>a length needing evaluation; array literals</p> <p>unary, index and call overloading</p> <p>transmute; float printing</p> <p>patterns, ranges, guards; a jump table</p> <p><code>#must</code>; a multi-result call in a return</p> <p>macros (W5)</p> <p><code>#expand</code> macros; inference through <code>Box(\$T)</code>; a length needing arithmetic</p> <p>a cross-file #run value; a Code value (declined)</p> <p>a chain <code>B :: A</code>; comparing types; Type as an annotation</p> <p>a per-frame line; inlined frames (none exist)</p> <p>a per-index <code>#no_abc</code>; any other build setting</p> <p>a cross-file or foreign proc value; comparing one</p> <p>cast to a pointer</p> <p>the difference <code>p - q</code>; <code>p[n]</code>; ordering</p> <p>hands out *u8 only; alignment; a growable region</p> |
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No error-handling model yet — ADR-0008 reserves the slot and the first half exists (several return values); #must is owed its own ADR. No GC and no RAIi, which is a design value rather than a gap. A union is untagged, so reading a field other than the one last written reinterprets bits: documented in three places because it cannot be diagnosed. No indirect calls, which is the single largest gap in the project and is what blocks the rest of W3.

## Compiler internals, 17 crates

| STAGE                             | STATE       | NOTE                                                             |
|-----------------------------------|-------------|------------------------------------------------------------------|
| Lexer, parser, CST, typed AST     | works       | Hand-written, error-recovering, trivia-preserving                |
| Formatter                         | works       | Pure function over the CST; lost source in 7 of the last 8 waves |
| HIR, name resolution, modules     | works       | Flat import merge; cycles legal; export filtering                |
| InternPool: types, values, layout | works       | One layout computation and one integer evaluator, shared         |
| Sema: signatures, checking        | works       | E0212 to E0257; no const-eval here, by design                    |
| MIR: typed SSA                    | works       | Block parameters, notphis; explicit bounds check and zeroing     |
| Mid-end                           | 5 passes    | Inline, forwarding, const-prop, DCE, plus the bounds-check strip |
| Const-eval                        | works       | Runs MIR through the bytecode VM                                 |
| VM: register bytecode, libffi     | works       | No JIT; indirect calls, and malloc from its own region           |
| Cranelift back end                | works       | Aggregate returns via sret; indirect calls via func_addr         |
| LLVM back end                     | not started | W8 owns it                                                       |
| Language server                   | 12 caps     | Diagnostics, hover, goto, completion, rename, actions, hints     |
| Neovim integration                | works       | Runtimepath dir, no plugin manager; 166 checks                   |
| Driver                            | stub        | Should consume the workspace notion that now exists              |

## Roadmap: W1 and W2 closed, W3 started

|                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>W1 Data</b>               | done | Numeric tower, wrapping ops, enum, enum_flags, union, arrays, views, cast, xx, operator overloading. Closed by ADR-0048.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>W2 Flow and scope</b>     | done | for with it and it_index, labelled break, defer, using, multiple returns, named and default arguments, #scope_module. Closed by ADR-0054; ADR-0055 then closed a gap six corpus files had carried for eleven waves.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>W3 Runtime core</b>       | done | context (ADR-0057), the bounds-check build setting (ADR-0058, finishing ADR-0003), indirect calls (ADR-0059), null plus a memory source (ADR-0060/0061), the allocator protocol (ADR-0062), push_context (ADR-0063), pointer arithmetic (ADR-0064), temporary storage (ADR-0065), and traps with backtraces (ADR-0066) — a trap names the frames that were live, byte-identically in both engines. Closed by ADR-0066; a source-level backtrace with inlined frames is deferred, because inlined frames have no runtime existence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>W4 Comptime</b>           | done | Delivered in sub-waves, because a 10-14 week wave cannot be verified the way a one-ADR wave can. All ten have shipped, and W4 is complete as scoped. A #run may call an imported procedure and appear in a body (ADR-0069), turning two internal compiler errors into working programs. An array length may name a constant (ADR-0070), which replaced the scheduled aggressive const folding after probing showed const-prop already did it. A type is a compile-time value (ADR-0071), which closed a silent miscompile — a type bound to a local compiled to an undefined value in a slot with no layout. Insert of a string literal lowers where it is written (ADR-0072), in the enclosing scope, every synthesized span pointing at the directive because jr-diag clamps an out-of-range offset rather than rejecting it. And a computed insert (ADR-0073) evaluates its operand at compile time and splices the text — the point sema and the VM become mutually recursive, PLAN section 5's named top risk, broken by an acyclic pre-pass that reuses the constant evaluator and re-lowers only the affected bodies rather than by salsa fixed-point recovery. An aggregate compile-time value (ADR-0074): a #run returning a struct or array interns as its element values rather than a byte image, because the pool is target-independent and an image is not. And type_info(T) (ADR-0075), reflection's first half: a type's kind, name, size and alignment, the numbers coming from the same layout_of every real layout decision uses, so reflection cannot disagree with the layout it describes. Type_Info is declared in modules/Basic in Jairs rather than inside the compiler, because it has to be spellable — no compiler-declared type can be named at all — and the resulting dependency on a declaration the compiler does not own is validated on lookup, so editing that struct is a diagnostic rather than a read of whatever now sits at the old offset. Getting there first needed a constant that may hold a string, which ADR-0074's own closing claim said was already done and was not: the fourth false scheduled dependency this project has found, and the first where the false claim was its own ADR about the very next wave. And Any (ADR-0076, ADR-0077), reflection's second half: any_of erases a value to a {Type_Info, *u8} pair and any_as reads it back, trapping unless the type matches — the erasing pointer conversion allowed only at that boundary, because a general one would make a wrong pointee type a silent wrong read. Nothing is reinterpreted, so neither back end needed a line. The checked read needed a runtime type identity the four-field Type_Info did not have — two type_info calls have different addresses, size and alignment collide, and name is unsound because a local and an imported type share a spelling — so Type_Info gained a stable id, the pool id the compiler already uses. And Type_Info's fixed-size per-kind facts (ADR-0078): a struct's field count and an array's or pointer's element type, added as flat s64 fields because a count is a number and an element type is a pool id — so they need none of the memory-ownership decision the variable-length field list does, which stays deferred. And #code (ADR-0080), unquoted source spliced into the enclosing scope — deliberately sugar over #insert, since #insert of a named constant already worked, so what it adds is no quoting and a body parsed where it is written. A Code value is declined rather than deferred: a quoted syntax tree is worth representing only once something can inspect or transform one. The same sub-wave refused a shipped silent miscompile found while probing (ADR-0079): a pointer or view inside a compile-time aggregate interned the evaluator's own address as an integer, giving 48 in one engine and a segfault in the other with no diagnostic. Out of W4's scope, each with a recorded reason: Type_Info's variable-length field list, which needs a declared static-data mechanism and is owed its own wave; a bare value coercing to Any, which needs a materialised temporary; and a #run reading another file's constant, which now reports itself rather than an internal error. |
| <b>W4.5 Pattern matching</b> | done | switch with exhaustiveness checking, a bare dot-member as a case (settling ADR-0041 §2 step 5), and a tagged variant type (ADR-0067, ADR-0068). Reordered ahead of W4 after checking showed its stated dependency on comptime was a want rather than a need. The variant follows ADR-0045 §1's own instruction — a different declaration form, not a change to union — and union is untouched, still untagged and still one word smaller.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>W5 Polymorphism</b>       | done | COMPLETE in fifteen sub-waves (ADR-0081 to ADR-0097). The last piece, #bake_arguments specialisation, lowers a declaration to a REAL procedure — a clone with the baked parameters dropped, their literals substituted and the kept ones remapped, which is the same machinery \$N instantiation uses, so W5 ends on a reuse rather than a new mechanism. A baked value must be a literal: ADR-0096 planned to use the const-eval pre-pass and building it showed that pre-pass runs AFTER lowering. #bake_arguments has its surface (ADR-0096) — a partial application producing a specialised procedure; its operand is a call so the named-argument spelling is the ordinary one, and its specialisation is refused E0276 pending the last W5 sub-wave. That refusal replaced a leaked gap report (the compiler could not lower main, please report it) which is right for an unknown gap and wrong for a named one. #modify is COMPLETE (ADR-0093/0094/0095): a compile-time predicate over an instantiation runs when a call binds the type variables, and a false refuses that instantiation (E0275) — so a template enforces its own constraints in code. It is hosted in file_mir, the only place with the expanded tree, its MIR and the VM, and rejections ride the existing diagnostics channel so it needed no new query. A predicate that fails to RUN is deliberately not a rejection. E0274 was retired when the predicate began running — the fourth by-design refusal raised then lifted. #modify has its surface (ADR-0093): a compile-time predicate over an instantiation, guarding a template in code rather than a comment. The block parses (the one procedure attribute                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

carrying a block) and formats with its body; a call is refused E0274 pending evaluation, because a parsed-and-ignored predicate would accept calls the author rejected — ADR-0058 rule for the third time. Its evaluation is designed and deferred. `type_info(T)` now reflects a BOUND type variable (ADR-0092) — a `$T` procedure can ask its own bound type size, field count or identity, each instantiation seeing its own. That was found missing while designing `#modify`, whose predicate needs exactly it, and fixing it also turned a sixth leaked internal error into working code. The `#expand SPLICE` works (ADR-0091): a call splices the macro body into the caller scope, so a macro can modify the caller local — deliberately unhygienic like `Jai`. A generated prelude binds each argument once (substituting per use would re-evaluate a side-effecting argument), and expression position gets a generated result local so one mechanism serves both. The MIR shows no calls at all. Refused by design: an early return (E0273), a void macro in expression position, a cross-file call (E0272, which had been reaching the VM as an internal error). `looks_like_proc_signature` needed `#expand` too — the token-set trap for the fifth time, since a void macro reaches neither arrow nor brace. `#expand` macros have their surface (ADR-0090): a macro parses, formats and checks like any procedure, and a call is refused E0272 pending the splice — a refusal that ships WITH the surface, because without it `#expand` was accepted and silently ignored (a macro behaved as an ordinary procedure). `jr_fmt` dropped `#expand` on the first run, caught by gate 5. `$N` comptime-value parameters are complete — surface, instantiation, and [N]T sized by one (ADR-0089), where two instantiations get genuinely different array types from one declaration. They work end to end (ADR-0087 surface, ADR-0088 build): `make :: ($N: s64)` called as `make(5)` evaluates the argument via the same acyclic pre-pass `#insert` uses, and appends a concrete procedure with `N` baked into the body — parameter list drops the `$N`, each reference to `N` becomes a literal. Two calls at the same value dedupe (ADR-0005 extended to values), distinct values instantiate separately. Mixed comptime+runtime params (scaled `:: ($N: s64, factor: s64)`) pass only the runtime one at the call site — a per-call arg-mask filters at MIR, teeth-checked (disabling it makes the verifier catch an arity mismatch). E0271 refuses a non-constant argument at the call's span. Before that, `$T` procedures work end to end (ADR-0081-0084): a `$T` parameter is inferred from the call — directly or through a pointer or view — instantiated once per distinct tuple of bound types, checked per instantiation, and run as an ordinary procedure in both engines, so nothing polymorphic survives to the back end. And polymorphic structs (ADR-0085, built per ADR-0086): `Box :: struct($T) { value: T; }` used as `Box(s64)` is a type constructor, and `Box(s64)` and `Box(bool)` are distinct types from one declaration with substituted fields and layouts, told apart in the pool by the type argument in the key the way `[2]s64` and `[3]s64` are. It changed the pool's most load-bearing invariant — a struct's identity was its declaration site — and was landed in two commits, a zero-behaviour-change representation refactor proven by an unchanged snapshot and test count, then the parameterised behaviour, so a half-built type-identity change could not hide a miscompile. W5 is done; next is W6 Metaprogram then W7 Stdlib (modify, `bake_arguments`, `expand`), and the deferred struct pieces (inference through `Box($T)`, using on one, cross-file, recursive `List($T)`), each a refusal today rather than a gap.

**W6 Metaprogram**  
**W7 Stdlib**  
**W8 Performance**

not started Workspaces, the compiler message loop, build scripts replacing makefiles, note attributes.  
not started Written in Jairs: String, dynamic array, hash table, Sort, Math, Random, File.  
not started LLVM via inkwell for release builds, inliner maturity, struct-of-arrays, SIMD, parallel sema. Also owns the compile-throughput number, still unpublished.  
not started Semantic tokens are the one LSP capability still missing; the other twelve landed early.  
not started Window creation through foreign calls, Metal then Vulkan, immediate-mode 2D.

**W9 Tooling depth**  
**W10 Graphics, in Jairs**

## What this session did, and how each wave was scoped

### FORTY-NINE WAVES SHIPPED

ADR-0049 through 0097: for and defer, using, aggregate returns, multiple returns, named and default arguments, scope visibility, imported constants, float constants, context, the bounds-check build setting, indirect calls, null plus a memory source, the allocator protocol, `push_context`, pointer arithmetic, temporary storage, trap backtraces, switch, tagged variants, compile-time run across files and in a body, an array length from a constant, a type as a compile-time value, insert of a literal and a computed string, an aggregate compile-time value, `type_info` and `Any`, code splicing, `$T` procedures, polymorphic structs, and `$N` comptime-value parameters with array lengths.

Test count 900 to 981. Corpus 116 to 193 files. Neovim checks 103 to 166. **W2, W3, W4.5 and W4 are all closed**, and **W5 is open** with six sub-waves shipped: `$T` procedures, polymorphic structs, and now `$N` comptime-value parameters and their instantiation. A `$N` call `make(5)` evaluates the argument via the same acyclic pre-pass `#insert` uses (ADR-0073), and appends a concrete procedure with the value baked into the body — the parameter list drops the `$Ns`, each reference to `N` becomes a literal. Two calls at the same value dedupe, distinct values instantiate separately, and mixed comptime+runtime params pass only the runtime ones at the call site. A per-call arg-mask filters at MIR, teeth-checked (disabling it makes the verifier catch an arity mismatch). W5 is complete. Next: W6 Metaprogram, then W7 Stdlib.

**Each wave was scoped by writing the feature and seeing what the compiler refused**, not by reading the handoff. That found gaps the handoff had missed in both directions: a proc-pointer struct field it called absent (it worked), and a void-returning proc-pointer type it never mentioned (unspellable, and blocking).

### THE THING WORTH YOUR DECISION

**Every wave is now committed as it greens**, on its own `feat/` branch, which closed the risk this box used to name. Nothing has been merged: `main` is still at `ec150a5`, and twenty-three waves sit on stacked branches ahead of it. That is a decision rather than a gap — merging needs your say-so, and the per-wave commits mean no work is at risk while it waits.

*The one open slice criterion is still a verified Linux x86-64 CI run, which needs a push.*

### TWO CLAIMS THE CODE DISPROVED

**An ADR corrected mid-wave.** ADR-0060 §4 asserted the VM dereferences a `malloc'd` pointer via `libffi`. Running the corpus file the same ADR promised would pass disproved it: the VM's memory is a linear region and a host address is not an offset into it. ADR-0061 corrects it — the VM allocates from its own region, native calls `libc`, the bits differ and nothing observes them.

**A diagnostic that could not be acted on.** Installing an imported `#foreign` procedure into an allocator field reported “expected (s64) -> \*u8, found (s64) -> \*u8” — the same text twice, because the types differ only in an invisible calling convention. It is E0256 now, the code that says to wrap it.